Electrostatically Driven Interactions between Hybrid DNA-Carbon Nanotubes

Xiangyun Qiu,[1,*] Constantine Y. Khripin,[2] Fuyou Ke,[1] Steven C. Howell,[1] and Ming Zheng[2]

[1]*Department of Physics, George Washington University, Washington, D.C. 20052, USA*

[2]*Polymers Division, National Institute of Standards and Technology, Gaithersburg, Maryland 20899, USA*

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Single-stranded DNA is able to wrap around single-wall carbon nanotubes (CNT) and form stable DNA-CNT hybrids that are highly soluble in solution. Here we report quantitative measurements and analysis of the interactions between DNA-CNT hybrids at low salts. Condensation of DNA-CNT hybrids by neutral osmolytes leads to liquid crystalline phases, and varying the osmotic pressure modulates the interhybrid distance that is determined by x-ray diffraction. Thus obtained force-distance dependencies of DNA-CNT hybrids show a remarkable resemblance to that of double-stranded DNA with differences that can be largely accounted for by their different diameters. This establishes their common physical nature of electrostatically driven interactions. Quantitative modeling further reveals the roles of hydration in mediating the interhybrid forces within the last nanometer of surface separation. This study also suggests the utility of osmotic pressure to control DNA-CNT assemblies at subnanometer precision.

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Hybridization of biomolecules and abiotic materials creates opportunities to design supramolecular complexes integrating the unique properties of the two distinct classes of materials. This has attracted widespread interest in understanding the design rules of biotic-abiotic hybridization and the forces governing the nanoscale assembly of such hybrids [1]. This study focuses on the hybrid of two important nanostructures of exceptional diversity—single-stranded DNA and single-wall carbon nanotubes (CNT), and reports quantitative measurement and analysis of the organizing forces between DNA-CNT hybrids.

Single-stranded DNA of lengths ranging from tens to hundreds of bases has been discovered to tightly wrap around CNTs and form stable DNA-CNT hybrids [2,3]. Single-wall CNTs are a group of single-atomic-layer-thick tubes with exceptional optical, electric, thermal, and mechanical properties [4]. The structure and property of each type of CNT are uniquely defined by its chirality (n, m), which denotes the folding vector when constructed by rolling up a graphene sheet. In terms of general mechanism, DNA-CNT hybridization is attributed to the amphiphilic nature of DNA, i.e., hydrophobic nucleobases and hydrophilic or charged phosphate backbone. As a result, the hydrophobic surface of CNT is buried by DNA bases and the hybrid solubility is conferred by charged DNA phosphates. While such a hybridization strategy works for wide ranges of amphiphilic biomolecules from bile salts to peptides and biopolymers [5,6], DNA exhibits exceptional capabilities such as attaining high solubility on the order of 20 mg/ml. Probably the most intriguing is the recognition between DNA of a specific sequence and CNTs of a specific chirality [2,3,7], the physical origin of which remains elusive. Importantly, this has enabled the sorting of CNTs by length and chirality via chromatography [3,7,8], producing single-chirality CNTs of highest purities to date. As a result, DNA-CNT hybrids overcome two major challenges of CNT technology, inherent heterogeneities of as-synthesized CNTs and refractory CNT-CNT clustering due to strong van der Waals (vdW) forces. With DNA being a versatile biomolecule, the DNA-CNT hybrid gives rise to a novel class of biotic-abiotic hybrids of unparalleled diversity for practical applications. It further allows fundamental studies of chirality-specific CNTs as amenable molecules and sequence-specific DNA on novel abiotic substrates.

DNA-CNT hybrids are rigid rodlike and highly charged molecules. They are usually prepared by sonicating aqueous suspensions of DNA and CNT mixtures, commonly referred to as the exfoliation process. The length of resultant CNTs typically ranges from 50 to 500 nm [9], much larger than their diameter (0.5–2 nm) and much smaller than their persistence length (>32 μm [10]). The hybrid's charge density is given by the bound DNA and has been measured via capillary electrophoresis and concentration assays to be comparable with that of double-stranded DNA [9,11]. Consequently, the anisotropic shape and polyelectrolyte nature of DNA-CNT hybrids have been widely exploited to assemble functional materials for applications in nanotechnology and nanomedicine. A few examples include their uses as aligned thin-film transistors [12], 2D CNT-based circuits [13], intracellular heavy metal ion sensors [14], and effective carriers for gene delivery [15]. Still, it remains challenging to attain high levels of CNT alignment or precise placement in programmed arrangements. Toward this goal, molecular self-assembly is actively sought after to create nanoscale structures via molecular manipulation [16]. However, while self-assembly seeks to organize molecules through intermolecular interactions without external direction, quantitative knowledge of the interhybrid interactions is missing due to the lack of direct measurements.

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Interactions between DNA-CNT hybrids are expected to be dominated by electrostatic forces between DNA strands and vdW forces between CNTs. Consequently, interhybrid interactions depend on DNA coverage density, salt condition, and CNT chirality. These distinct dependencies also make DNA-CNT hybrids a unique system to investigate DNA-dominated electrostatics in an unfamiliar environment, as well as CNT-dominated vdW forces that may become possible to probe owing to the conferred solubility. While there are currently a number of first-principle and empirical theories of both electrostatics and vdW forces [17–19], accurate theoretical calculations are still difficult. For example, electrostatic interactions in aqueous environment involve many ubiquitous ion and water molecules, and detailed descriptions of the multicomponent systems are challenging [19]. Extensive studies on well-defined systems such as double-stranded DNA (dsDNA) have yet to fully explain experimental observations of inter-DNA forces [20]. Meanwhile, in contrast to the availability of theoretical methods, measuring the vdW forces between CNTs has proven to be a formidable task [17]. Quantitative measurement is thus critically needed in order to understand and control the multifactorial interactions between DNA-CNT hybrids.

We have explored the use of osmotic stress method (OSM) in conjunction with x-ray diffraction (XRD) to measure and understand interhybrid interactions of DNA-CNTs at low salts. Owing to the formation of liquid crystalline DNA-CNT phases under external osmotic pressure, the interhybrid distances were determined by XRD at varied external pressures, giving the force-distance dependences of DNA-CNT hybrids. By direct comparison with the forces between dsDNA strands measured under the same conditions, interhybrid forces are shown to be driven by electrostatics. Theoretical modeling further reveals the necessity to consider the restructuring of the hydration shells upon close approaching, giving rise to the form of hydration forces within the last nanometer of surface separations.

Preparation of DNA-CNT hybrids followed the protocols established by one of our labs as in Ref. [2]. Single-stranded $(GT)_{20}$ DNA was obtained from Integrated DNA Technologies and chosen for its strong binding to CNT [2]. Single-wall CNT (CoMoCAT SG65) was obtained from SouthWest NanoTechnologies and chosen for its high enrichment of (6, 5) chirality CNTs ($> 50\%$). Other chemicals were obtained from Sigma and used as received. In brief, CNTs in powder form are briefly sonicated in deionized water before adding to a solution with equal weight of $(GT)_{20}$ DNA. The mixture (DNA-CNT of ~ 2 mg/ml) is further sonicated in ice bath for 2 hours, followed by centrifugation to remove impurities and undispersed CNTs. No precipitate can be observed after additional centrifugation at 20000 g for 16 h. Thus prepared DNA-CNTs have been shown to be free of CNT bundles by atomic force microscopy [2]. We further examined the sample quality using UV-vis-nIR and solution small angle x-ray scattering (see Supplemental Material [21], Fig. S1). XRD measurements of liquid crystalline DNA-CNT phases were carried out at 20 °C with an in-house instrument which integrates a microfocus fixed Cu-anode source and an image plate detector.

OSM is an ensemble method to quantify intermolecular forces at surface separations of the last several nanometers that are important for molecular recognition and assembly [22]. OSM explores the use of neutral osmolytes to force molecules to approach each other, and its physical origin can be understood in terms of molecular crowding or depletion force [22]. As illustrated in Fig. 1(b), osmolyte-induced condensation of DNA-CNT hybrids leads to coexistence of two phases (i.e., the DNA-CNT phase and the osmolyte phase) between which solvent and ions can exchange freely. Such samples typically consist of ~ 200 μg DNA-CNT phase (pellet) in 1 ml of osmolyte phase (solution) and are equilibrated over two weeks with one change of the bathing osmolyte solution. As phase equilibrium is reached, the pressure (or force) between

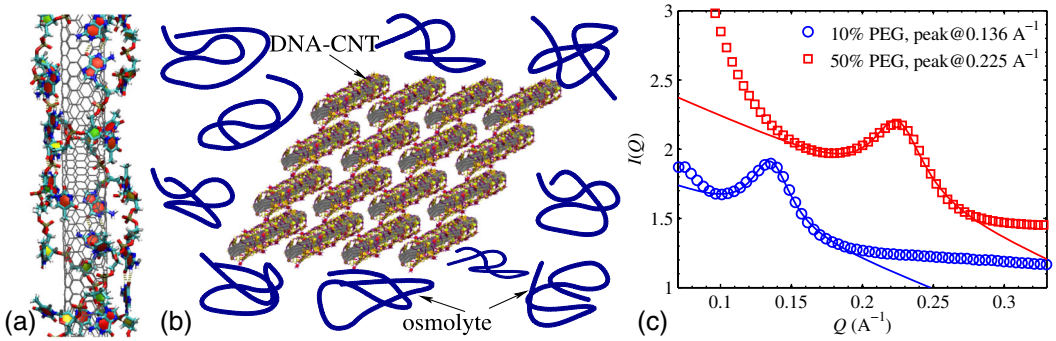


FIG. 1 (color online). Osmolyte-driven self-assembly of DNA-CNT hybrids and structure characterization. (a) One proposed DNA-CNT structure, the β-barrel DNA helical motif around a (6, 5) CNT [30]. (b) Cartoon illustration of the osmotic stress method showing coexistence of the assembled DNA-CNT phase and the osmolyte phase. (c) XRD profiles (symbols) of the DNA-CNT phase equilibrated against 10% and 50% PEG8k (weight/weight) in 150 mM NaCl 1 × TE pH 7.5, respectively. Error bars of Q and $I(Q)$ are smaller than the sizes of symbols. Here $Q = 4\pi \sin(\theta)/\lambda$ is the scattering vector, where 2θ is the scattering angle and λ is the x-ray wavelength. Both peaks are fit with Lorentzian functions (solid lines) giving comparable peak widths of 0.033 Å^{-1} (after correction by instrument resolution of 0.006 Å^{-1}).

DNA-CNTs is balanced by the osmolyte pressure known from the osmolyte concentration. Provided that the interhybrid distance can be determined independently, OSM measurements at varied osmolyte concentrations give the interhybrid force as a function of the interhybrid distance.

Structure ordering of DNA-CNTs is a key requirement for XRD to determine their interhybrid distances to Angstrom resolution. The formation of liquid crystalline (LC) phases was first examined with polarized light microscopy [see Supplemental Material [21], Fig. S2(a)]. While the appearance of LC phases has been observed for the hybrids of CNTs with molecules such as genomic DNA and chitosan [23–25], it has not been shown for the hybrid of CNT and oligomeric single-stranded DNA, and, most importantly, no XRD studies of their nanoscale structures have been performed to our knowledge. Here we report the presence of well-defined diffraction peaks shown in Fig. 1(c) [see Supplemental Material [21], Fig. S2(b) for raw XRD intensities], indicating the occurrence of positional ordering of the DNA-CNT LC phase at high densities. Under 10% weight/weight polyethylene-glycol 8000 Dalton (PEG8k) in 150 mM NaCl, the peak position is at $Q = 0.136$ Å^{-1}, giving a Bragg spacing of 46.2 Å and an interhybrid distance of $d = 46.2 \times 2/\sqrt{3} = 53.3$ Å. Note that the $2/\sqrt{3}$ factor is applied to account for the local hexagonal packing that has been observed for polydisperse rods at close separations [26] (see Supplemental Material [21], Fig. S3, for detailed discussions). Presuming a DNA-CNT diameter of ~28 Å (discussed in the Supplemental Material [21], Fig. S1), this indicates substantial "room" (~ 25 Å) between neighboring DNA-CNT surfaces, likely arising from strong interhybrid repulsion at low salt. This room is expected to be reduced by increasing external osmotic pressure. Accordingly, the peak position under 50% PEG8k shifts to a much higher Q of 0.225 Å^{-1} and gives $d = 32.2$ Å, 21.1 Å closer than under 10% PEG8k. It is worth noting that osmotic pressure can thus be a very practical tool for controlling CNT assemblies with high precision. In addition, the widths of XRD peaks provide information on the coherent length of positional ordering. Both peaks in Fig. 1(c) and the peaks under other [PEG8k]s between 10% and 50% (not shown) give a comparable full width at half maximum of $\sigma = 0.033$ Å^{-1} which corresponds to ~188 Å coherent length of positional ordering based on the Scherrer formula $D = 2\pi/\sigma$ [27]. This indicates that the self-assembled DNA-CNTs lose positional ordering just over a few interhybrid distances, likely explaining the absence of higher order XRD peaks. In comparison, PEG8k-condensed dsDNA arrays (long genomic type) give XRD peaks of widths ~0.008 Å^{-1}, about four times sharper than the DNA-CNT XRD peak. Possible causes of such short-range positional ordering of DNA-CNTs include its fast nucleation kinetics under studied conditions and their high rigidity combined with large aspect ratio restricting structural relaxation.

By varying the PEG8k concentration, the force-distance relations of DNA-CNTs are obtained and shown in Fig. 2 in varied salt conditions. A monotonically increasing repulsive force is observed upon decreasing the interhybrid distance. As the CNT vdW forces are attractive, the measured repulsion between DNA-CNTs suggests the dominance of the electrostatic interaction arising from the charged DNA. The expected electrostatic nature is in qualitative agreement with the observation in Fig. 2(a) that stronger repulsions are measured in 50 mM NaCl than in 150 mM NaCl at the same inter-hybrid distances. Efforts to probe wider parameter space (e.g., ion valence and concentration) are underway as various strategies are pursued to lower sample consumption. As this is the first measurement of the force-distance dependences of DNA-CNT hybrids to our knowledge, we have verified that the two weeks of incubation time in osmolyte solution is sufficient to reach phase equilibrium (see Supplemental Material [21], Fig. S4). Given the relatively large interhybrid distances (32–70 Å), the slight diameter spread of the CoMoCAT SG65 CNTs (± 1 Å) is considered insignificant, noting that studying single-chirality CNTs is currently limited by the high cost and low yield of their purification [3]. As thus prepared, DNA-CNT hybrids have lengths ranging from 50 to 500 nm [9], we have further established that the measured interhybrid distances are independent of the lengths of DNA-CNT hybrids (see Supplemental Material [21], Fig. S5).

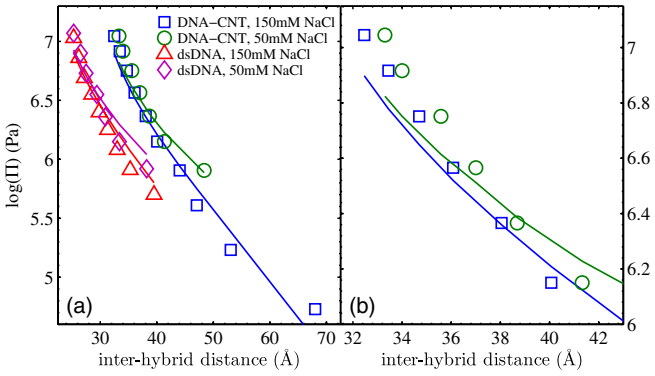


FIG. 2 (color online). Measured force-distance curves of DNA-CNT hybrids and calculations based on electrostatic theories. The force is given by the osmotic pressure Π in Pascal. Uncertainties for inter-hybrid distances range between 0.15 and 0.35 Å. The concentrations of PEG8k were measured by weight to accuracy better than 0.1% and calculations of pressure Π used well-established parameters [22]. Both error bars are thus smaller than symbols. In (a), as-measured experimental data are shown as symbols. Solid lines show the calculated force-distance curves using the cylindrical cell model with the DNA-CNT's diameter taken as 28 Å. Note that the rightmost data point for each curve represents the lowest PEG8k concentration needed to condense DNA-CNTs, e.g., 20% PEG8k (Π ~ 8 atm.) in 50 mM NaCl and 5% (Π ~ 0.5 atm.) in 150 mM NaCl. In (b), the high-pressure (i.e., short-distance) region is zoomed in to illustrate the crossovers between experimental data and theoretical calculations, illustrating the different slopes between experiment and theory.

To establish the predominant electrostatic nature of interhybrid interactions, we next compare measured forces with theoretical calculations. Using a linear charge density of -0.6 e/nm derived from the 1.2:1 ratio of DNA:CNT by weight, recently determined by Khripin *et al.* [9], we calculate the electrostatic forces by numerically solving the nonlinear Poisson-Boltzmann (NLPB) equation with the cylindrical cell model (CCM) of rodlike polyelectrolytes [28]. Each CCM calculation considers a cylindrical cell whose dimension gives the interhybrid distance, with a charged rod of known diameter and charge density placed in the center. With the additional condition of zero electric field at the cell outer boundary, theNLPB equation is solved and the electrostatic force between two rods is given by the osmotic pressure of the ions at the cell boundary in excess of the ion osmotic pressure in the bulk solution. As shown in Fig. 2(a), measured and calculated forces agree fairly well on the overall magnitudes and overall trends in both 150 and 50 mM NaCl, lending strong support to the nature of dominating electrostatic interactions and the insignificance of CNT vdW forces. For further verification, measurements under identical salts were carried out for dsDNA (of genomic origin) with practically identical linear charge density as the DNA-CNT. The results are shown in Fig. 2(a) together with CCM calculations for which the only different parameter is the dsDNA diameter of 20 Å (compared to 28 Å for DNA-CNT). For both DNA-CNT and dsDNA, the CCM calculations without fitting parameters appear to explain the experimental data in terms of general behaviors. However, close examinations, especially in the high–pressure region, reveal crossover patterns between experimental and theoretical curves for both DNA-CNT and dsDNA. As shown in Fig. 2(b), the CCM slopes are correlated with the salt conditions qualitatively via the Debye screening length as expected, whereas the experimental slopes appear independent of the salt condition for both DNA-CNT and dsDNA. This constitutes a significant discrepancy we discuss later. In short, the force-distance dependences of DNA-CNT hybrids show close resemblances to that of dsDNA at low salts, establishing their predominant electrostatic nature. However, measurements are not completely described by basic theoretical models of electrostatics.

To further probe the nature of DNA-CNT interactions, we then shift the dsDNA curves along the x axis by 8 Å to account for their difference in diameter. This leads to a surprisingly good match between the DNA-CNT and dsDNA curves as shown in Fig. 3(a). While there may exist quantitative differences (see Supplemental Material [21], Fig. S6 for details), it is remarkable that their force-surface separation dependences show essentially identical forms. It also becomes more evident that the high-pressure regions for all curves show indistinguishable slopes. Such salt independent behaviors at high pressure have been observed previously for dsDNA as well as other molecules and can be explained by the formulation of hydration forces that become important at close spacings [22].

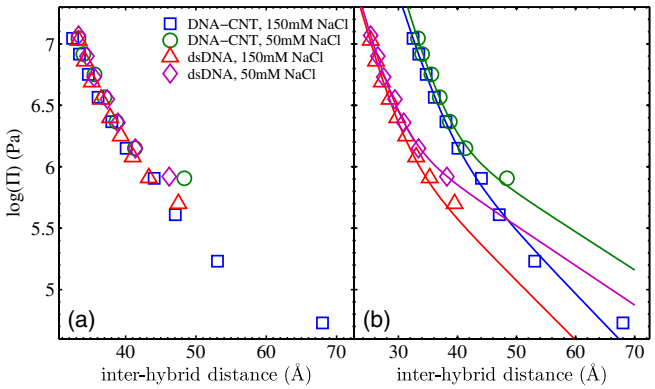


FIG. 3 (color online). Measured force-distance curves of DNA-CNT and dsDNA and fits based on the hydration force formalism. In (a), the same experimental data as in Fig. 2(a) are shown, except that the curves for dsDNA are offset by adding 8 Å to the interhybrid distance. In (b), symbols show the same data as in Fig. 2(a); solid lines show the fits using two exponentials including the hydration and electrostatic forces.

The hydration force originates from the "bound" hydration shells around molecules that can extend into nanometer ranges and come into contact upon the approaching of molecular surfaces. The consequent restructuring of hydration shells incurs substantial and usually dominant energetic cost at the last nanometer of separation, giving rise to universal hydration forces at short range. One signature of such force is its exponential form with a decay length of ~ 3 Å reflecting the water correlation length [22]. We then fit the measured force-distance curves with the sum of two exponentials, one short-range hydration force with decay length of ~ 3 Å and one electrostatic exponential force with the Debye length (13.6 Å for 50 mM NaCl and 7.8 Å for 150 mM NaCl) as its decay length. Such fits are shown in Fig. 3(b) and excellent agreements are obtained for all curves, quantitatively corroborating the nature of hydration forces at short range. Importantly, combining the hydration force and the electrostatic force allows us to fully parametrize their interaction forces under wide-ranging conditions, which can guide practical applications and theoretical developments.

Rather than suggesting the diminishing role of electrostatics, the prominent hydration forces at short range arise from the high charge density of DNA-CNTs (and dsDNA as well). More generally, charged molecular surfaces interact strongly with polar solvent (i.e., water) and ions. Qualitatively, this leads to salt-independent (i.e., hydration) and salt-dependent (i.e., electrostatic) forces. Both forces are modulated by molecular charges and are thus difficult to separate, which makes it challenging to quantitatively predict electrostatics in an aqueous environment. On the other hand, the complete absence of CNT vdW forces is worth noting. A number of theoretical models have been proposed to compute vdW forces of graphitic materials, e.g., the continuum Lennard-Jones model by integrating the C-C pair potentials [18] and the Hamaker coefficient calculation

based on one-electron dielectric response functions [29]. Using the value of the Hamaker coefficient of ~700 zJ in medium and Eq. (4) from Ref. [29], we estimate an attractive CNT force of 0.24 pN per 10 Å length for CNTs when separated by aqueous medium of thickness $t = 30$ Å. This gives an equivalence of 0.6 atm. "negative pressure" at interhybrid distance of 41 Å [using 11 Å as the vdW outer diameter of a (6, 5) CNT], while the measured pressure is ~10 atm. (see Fig. 3). Therefore, the CNT vdW force may be present but dominated by the electrostatically driven force of DNA. Furthermore, the vdW attraction between parallel CNTs increases upon approaching in the form of $1/t^{5/2}$, much slower than the exponential hydration force (decay length ~3 Å) in the measurement range down to $t = 20$ Å [29]. In order to probe the role of CNT vdW forces, the predominant electrostatic contribution from DNA needs to be weakened, which would also weaken the level of associated hydration and subsequently the hydration forces at short ranges. This may be achieved by reducing the DNA charge density or increasing the salt condition. For example, our work in progress has observed the condensation of DNA-CNT in divalent salts under which dsDNA remains soluble, suggesting the nonnegligible role of CNT vdW forces, though further studies are needed to rule out the wrapped DNA as the origin of the observed attraction in divalent salts. In addition, electrostatic and vdW forces have qualitatively different dependencies on parameters such as CNT chirality, DNA density, and salt condition; this may allow the decomposition of the two forces through systematic measurements.

In summary, we quantified the interactions between DNA-CNT hybrids and established the common physical origins of DNA-CNT and dsDNA forces at low salts. Wrapping flexible single-stranded DNA around rigid CNTs is observed to lead to electrostatic interactions similar to semirigid dsDNA, suggesting the possible use of CNT as a substrate to study its dispersing molecules (e.g., peptides). Meanwhile, the methodologies used here can be applied generally to other types of CNT hybrids and the measured force-distance relations likely promote both fundamental understanding and practical self-assembly of amenable CNT hybrids.

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*xqiu@gwu.edu

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Supplemental Data for **"**Electrostatics-driven Interactions between Hybrids of DNA-Carbon Nanotubes"

Xiangyun Qiu[1,*], Constantine Khripin[2], Fuyou Ke[1], Steven C Howell[1], Ming Zheng[2]

[1]Department of Physics, George Washington University, Washington, DC 20052

[2]Polymers Division, National Institute of Standards and Technology, Gaithersburg, MD 20899

**email: xqiu@gwu.edu*

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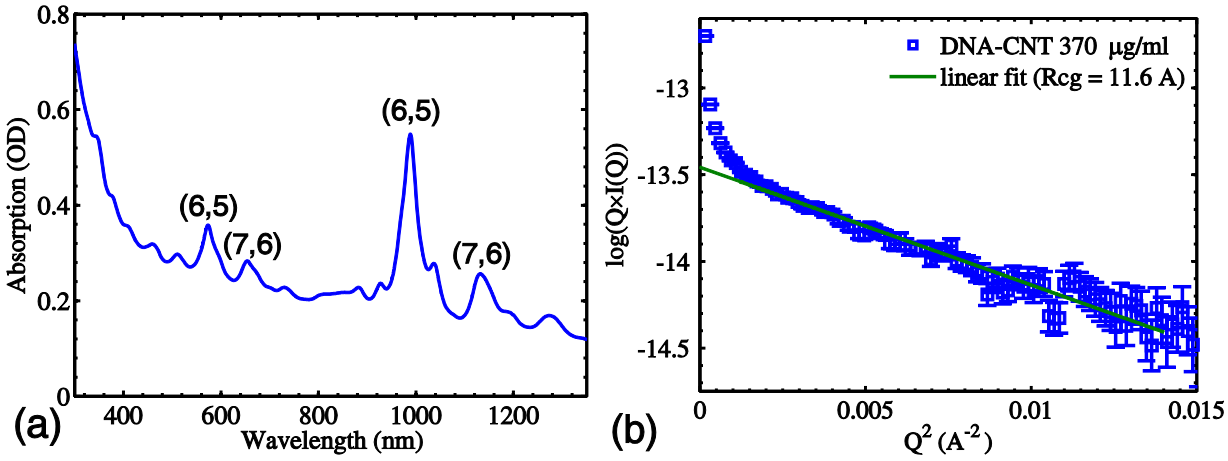


Figure S1. Characterization of DNA-CNT dispersions. (a) UV-vis-nIR spectrum of the DNA-CNT stock solution. The major peaks are labeled by the specific CNT chirality (n,m). The primary peak around 990 nm corresponds to the E_{11} optical transition of (6,5) CNT arising from the Van Hove singularities in its electron density of states. (b) Solution small angle x-ray scattering (SAXS) profile (symbol) of DNA-CNT solution shown in the form of log(Q×I(Q)) vs. Q^2 to obtain its cross-sectional radius of gyration R_{cg}. Here $Q=4\pi\sin(\theta)/\lambda$ is the scattering vector, where 2θ is the scattering angle and λ is the x-ray wavelength. SAXS measurements of DNA-CNT solutions were carried out at the G1 station at the Cornell High Energy Synchrotron Source (CHESS), Ithaca, NY. For a rod-like molecule with length L, its R_{cg} is determined by the slope of a linear region in an intermediate Q range of $1/L < Q < 1/R_{cg,}$ which gives 0.002 $Å^{-1} < Q < 0.1$ $Å^{-1}$ for L=50 nm and R_{cg}=10 Å [1]. A linear regime is evident at $Q > 0.045$ $Å^{-1}$, and fitting (shown as solid line) gives R_{cg} of 11.6(3) Å. This confirms not only the rod-like structure of DNA-CNT hybrids but also the highly uniform diameters of fully dispersed CNTs (i.e., free of multi-CNT bundles). In addition, we would like to discuss the following observations. (1) The rise of

log(Q×I(Q)) upon Q approaching zero is suggestive of inter-hybrid associations that are likely dynamic in nature because our sample purification steps remove static agglomerates if present. We attribute this low-Q upturn to the entanglement of DNA-CNT hybrids of 50-500 nm lengths at the studied concentration of 370 μg/ml, e.g., the overlapping concentration is 170 μg/ml for an average length 100 nm. This also offers an explanation to the observation of a linear regime in log(Q×I(Q)) vs. Q^2 starting from a larger Q than expected (0.045 vs. 0.002 $Å^{-1}$). (2) It is non-trivial to determine the steric diameter of DNA-CNT hybrids from R_{cg} because the hybrid cannot be taken as a uniform cylinder and the R_{cg} from SAXS is weighted by all atoms' electron-density contrasts in aqueous solutions. For example, the measurement of a 50 base-pair dsDNA construct gives R_{cg} of 8.3 Å, while its diameter is 20 Å rather than a value of 2x8.3*√2=23.5 Å for a uniform solid cylinder. An estimate of the DNA-CNT diameter by assuming the same scaling as dsDNA gives a value of 11.6/8.3*20=28 Å, which is coincidently close to the sum of the dsDNA diameter of 20 Å and the (6,5) CNT diameter of 7.5 Å.

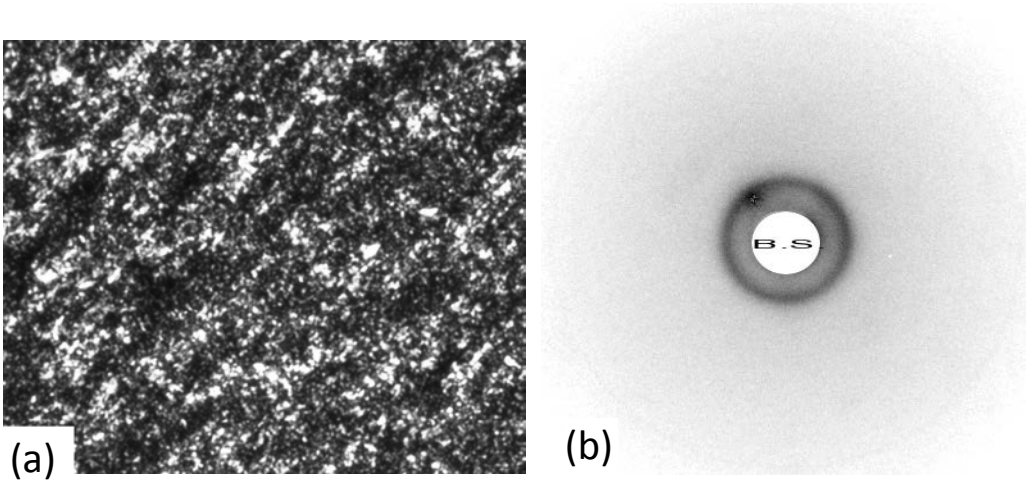


Figure S2. Macroscopic and microscopic characterization of osmolyte-induced liquid crystalline phases of DNA-CNT hybrids. (a) Polarization light microscopy of condensed DNA-CNTs in 150 mM NaCl and 10% (weight/weight) polyethylene-glycol 8000 Dalton (PEG8k). It shows marked birefringence indicating the formation of liquid crystalline (LC) phase, as expected from its rod-like shape. (b) Raw x-ray diffraction (XRD) 2D intensities of LC DNA-CNTs in 10% PEG8k. The dark spot to the upper-left of beam stop is an artifact due to the circular scanning of the image plate detector. The XRD peaks are relatively broad (full width at half maximum ~0.033 $Å^{-1}$), and no higher order XRD peaks were observed.

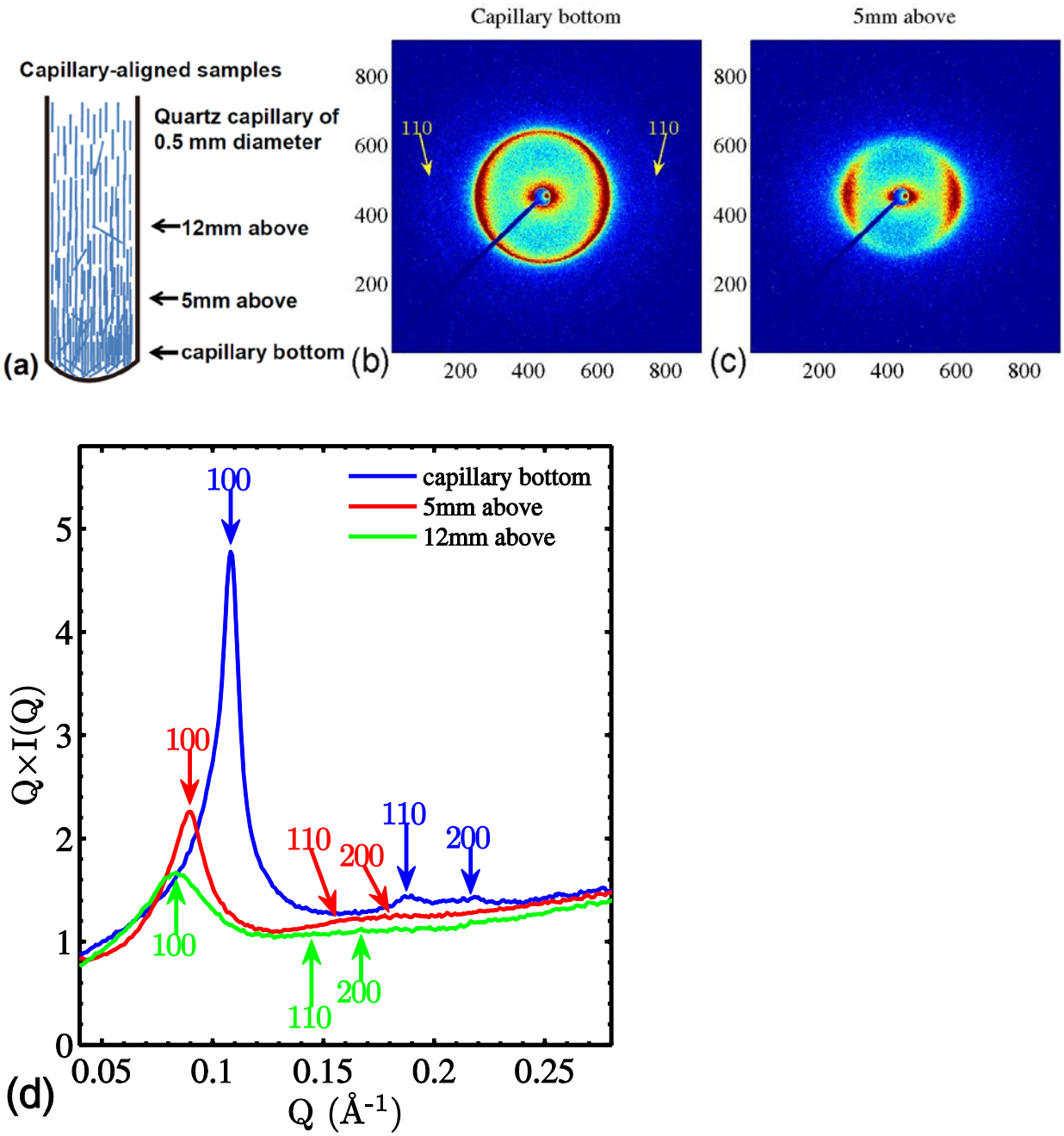


Figure S3. Experimental observations in support of local hexagonal packing of the PEG8k-condensed samples. Here we present evidences of higher order 110 and 200 XRD peaks from capillary-aligned samples that are consistent with hexagonal packing. These samples have much improved ordering than the PEG8k-condensed samples, which enables the observation of higher order XRD peaks. (a) Cartoon illustration of the capillary-aligned samples. In brief, these samples are prepared by loading concentrated DNA-CNT solutions into a 0.5-mm-diameter

quartz capillary and then centrifuging for 24-48 hours at low-medium speeds, during which the shearing force aligns the rigid hybrids along the capillary axis. This method results in varying gradients of hybrid density and alignment along the capillary axis as illustrated. The main objective of this experiment is to prepare macroscopically aligned hybrids at nanometer separations for research (e.g., structure determination via fiber diffraction) and potential applications. Note that the samples in the capillary are not in equilibrium and will slowly homogenize into a uniform DNA-CNT solution. (b) & (c) The XRD patterns from the samples at the “capillary bottom” and “5 mm above” locations, respectively. The salt condition is 150 mM NaCl in 1×TE buffer without PEG. Macroscopic alignments are evidenced from the anisotropic XRD rings observed, noting that the ‘capillary bottom” location, though with the highest packing density and sharpest XRD peak, usually shows lower degree of macroscopic alignment possibly due to the “collapse” of hybrids at the bottom. It is worth noting that a faint 110 reflection ring can be discerned in (b). (d) The integrated scattering profiles (Q×I(Q) vs. Q) at three locations along the capillary as labeled in (a). Pronounced 100 XRD peaks are observed and their positions shift to lower Q as expected from the decreasing hybrid density from bottom up. The 100 peaks also broaden when moving up along the capillary. For the “capillary bottom” sample with the sharpest 100 peak (width ~0.009 $Å^{-1}$), higher order peaks are observed in consistency with the 110 and 200 reflections of a hexagonal lattice, though they are rather weak as a result of improved but still poor positional ordering. For the “5mm above” sample with a 100 peak width of 0.018 $Å^{-1}$, a very broad feature is still identifiable as overlapping 110 and 200 peaks. For the “12mm above” sample with a 100 peak width of 0.033 $Å^{-1}$, higher order peaks are no longer visible. Overall, these observations establish local hexagonal positional ordering of DNA-CNT hybrids, and further indicate that highly ordered samples are needed to observe the higher order peaks. The broad widths and high-Q positions of the XRD peaks from PEG-condensed hybrids likely contribute to the difficulty in observing their higher order peaks.

In sum, the presence of higher order XRD peaks from the capillary-aligned samples strongly supports local hexagonal packing of DNA-CNT hybrids at close separations. This led us to account for the hexagonal correction factor of $2/\sqrt{3}$ to compute inter-hybrid distances from Bragg spacings. The main reason for using inter-hybrid distances is that the physical distance between hybrids, rather than the Bragg spacing referring to the distance between Bragg diffraction planes, is preferred to define the inter-hybrid force-distance relationships.

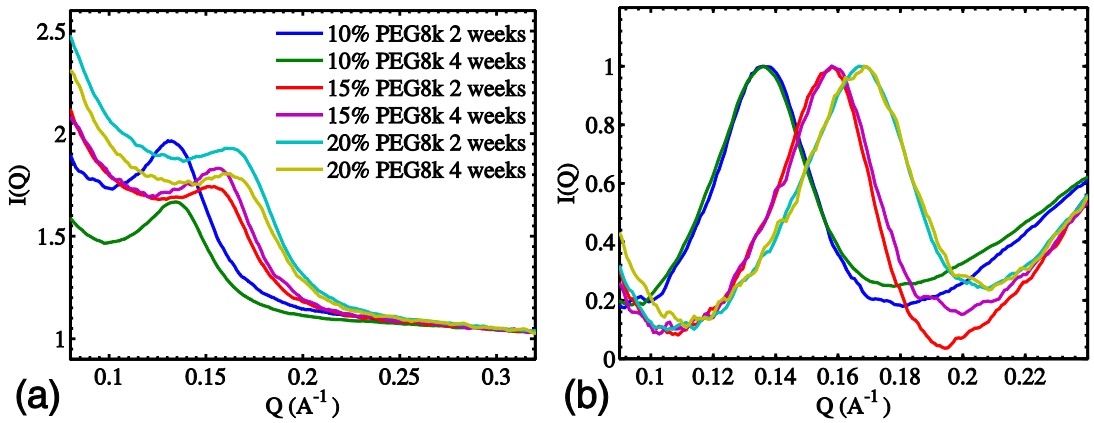


Figure S4. The positions of XRD peaks are independent of the incubation time in PEG8k solutions. (a) Shown are the XRD peaks from three samples incubated for two and four weeks respectively. The peak position and shape are essentially the same between two and four weeks. Data have slightly varying levels of scattering background because of the varied levels of pellet packing fractions when manually loading our x-ray sample cell, leading to different signal to background ratios. (b) Shown are the same XRD peaks in (a) with a linear background subtracted, indicating identical peak profiles and equilibration-time independence of inter-hybrid distances.

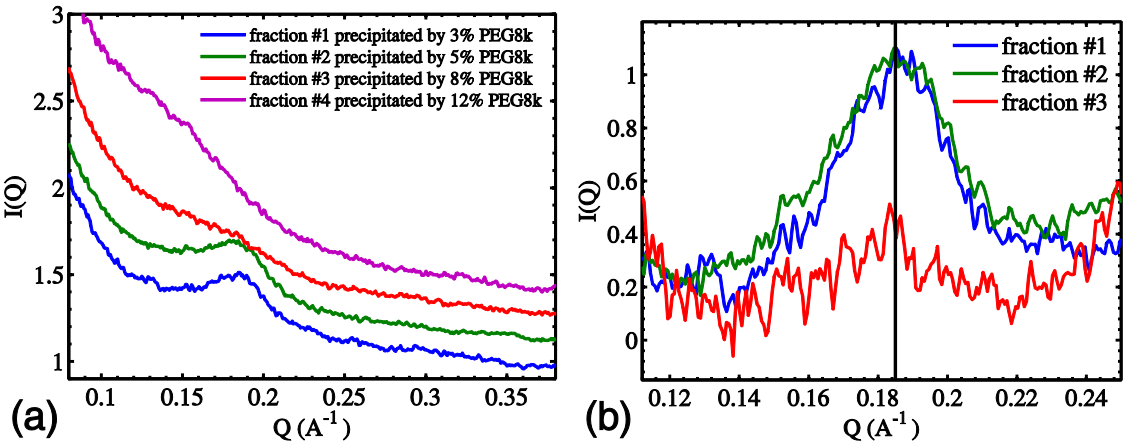


Figure S5. The positions of XRD peaks are independent of the length of DNA-CNT hybrids. For the DNA-CNT stock with 50-500 nm length distribution, we followed Ref. [2] to fractionate the tubes by length using PEG8k of various concentrations in ~400 mM NaCl salt. The strategy is to preferentially precipitate long tubes at certain low PEG8k concentration that is increased step-wise. In the mean time, at each PEG concentration, the precipitate is removed and taken as one fraction. (a) Shown are the XRD peaks from the four fractions step-precipitated by PEG8k at

concentrations indicated by the legend, noting that 12% PEG8k precipitates DNA-CNTs of all lengths. As the average length of DNA-CNT decreases from fraction #1 to #4, it is observed that long DNA-CNT hybrids give better defined XRD peaks, while the shortest DNA-CNTs (fraction #4) does not show any noticeable XRD peak. (b) Shown are the same XRD peaks in (a) for fractions #1-3 with a linear background subtracted, indicating consistent peak positions for DNA-CNT hybrids of different lengths. Atomic force microscopy experiments in Ref. [3] showed that short DNA-CNT hybrids tend to have more structural defects. Our observation further suggests that such defects also affect the hybrid's ability to form well-ordered structures, however, the inter-hybrid distances of the ordered structures, if formed, are not affected. We speculate that the DNA-CNT condensates may consist of two phases, one ordered phase (mainly long tubes) and one disordered phase (mainly short tubes). Above all, this result shows that our inter-hybrid distance measurements (via XRD peak positions) are independent of the length of DNA-CNTs.

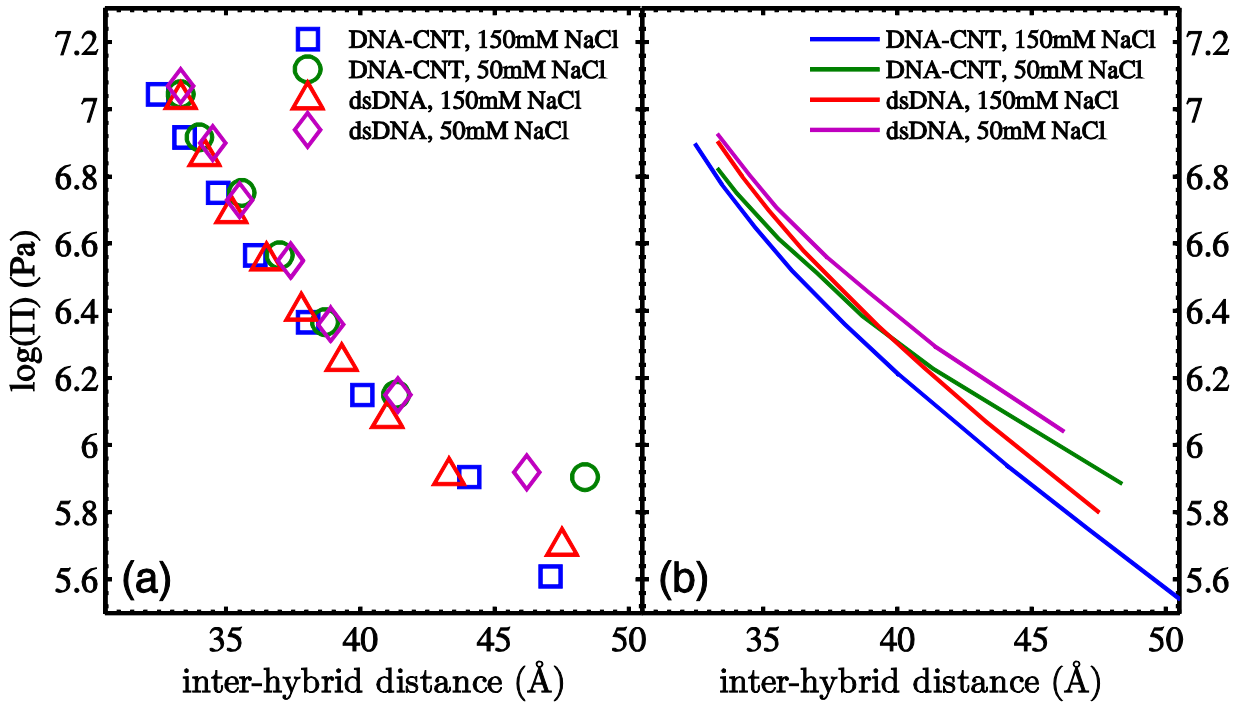


Figure S6. Measured force-distance curves of DNA-CNT and dsDNA and calculations based on the cylindrical cell model (CCM). In (a), shown are the high pressure regions of the same experimental data as in Fig. 3a (i.e., the curves for dsDNA are offset by adding 8 Å to inter-hybrid distance). In (b), shown are the high pressure regions of the same CCM curves as in Fig. 2a, except that now the inter-hybrid distances for dsDNA are offset by adding 8 Å. The purposes are two-fold. The first is to show the excellent match between the force-surface

separation curves for DNA-CNT and dsDNA especially at medium to long inter-hybrid distances, though very small differences may exist at the highest pressure region where the dsDNA may give slightly higher pressures. If true, this is likely physically significant because the origin is possibly the CNT van der Waals force that slightly lowers the force between DNA-CNTs compared with dsDNA. We are working to better define the high pressure region so as to reliably determine their differences, which are likely the CNT van der Waals forces. The second is to show that the CCM calculations do not match as closely as the experimental curves. This is as expected because DNA-CNT and dsDNA, though with identical linear charge density, have different diameters (28 vs. 20 Å) and thus different surface charge densities. The curves in (b) show that dsDNA gives larger forces at the same surface separations than DNA-CNT, consistent with the higher surface charge density of dsDNA (~140% of that of DNA-CNT).

References

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[2] C. Y. Khripin, N. Arnold-Medabalimi, and M. Zheng, Acs Nano **5**, 8258 (2011).

[3] C. Y. Khripin *et al.*, Anal Chem **84**, 8733 (2012).